Umar Habib

B.S. Computer Science (Senior) — LUMS, SSE
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EDUCATION

Lahore University of Management Sciences (LUMS) **2023 – 2027**
B.S. Computer Science with a Minor in Robotics CGPA: 3.65 · Major GPA: 3.83
Dean's Honour List (2024–25, 2025–26)

RESEARCH & TECHNICAL INTERESTS

Networks · Security & Privacy · Distributed Systems · Robotics

PUBLICATIONS

- **Made with AI?: A First Large-Scale Measurement of AI Disclosure Labels on YouTube**
  Muhammad Abubakar Mughal, Muhammad Muiz Farhan, **Umar Habib**, Ihsan Ayyub Qazi, Zafar Ayyub Qazi
  Under review at the ACM Web Conference (WWW), 2027.
- **Hive-Nav: Belief-Guided Multi-Robot Room Classification and Open-Vocabulary Object Search**
  Muhammad Hussain Habib Chaudhry*, **Umar Habib***, Hassan Jaleel. (* *equal contribution*)
  Under review at the IEEE International Conference on Robotics and Automation (ICRA), 2027.

EXPERIENCE

Research Assistant - Distributed and AI Systems (DAIS) Lab **Nov. 2025 – Present**
Lahore University of Management Sciences (LUMS)
Supervisor: Dr. Zafar Ayyub Qazi

- Co-authored *Made with AI?*, now under review at the ACM Web Conference (WWW) 2027.

Teaching Assistant - CS 5803: Applied Cryptography **Aug. 2026 – Present**
Lahore University of Management Sciences (LUMS)
Instructor: Dr. Muhammad Ali Siddiqi

- Selected on the basis of prior performance in the course.
- Held regular office hours to support students with course concepts and assignments.
- Graded all quizzes and assignments, and invigilated all quizzes and exams.

Teaching Assistant - CS 100: Computational Problem Solving **Jun. – Jul. 2026**
Lahore University of Management Sciences (LUMS)
Instructor: Dr. Malik Jahan Khan

- Designed several lab exercises and quizzes used in the course.
- Held regular office hours and supervised twice-weekly lab sessions.
- Graded labs as part of the TA team, and invigilated all quizzes and exams.

Teaching Assistant - CS 382: Network-Centric Computing **Jan. – May 2026**
Lahore University of Management Sciences (LUMS)
Instructor: Dr. Zartash Afzal Uzmi

- Selected on the basis of prior performance in the course.
- Held regular office hours and conducted tutorials for students.
- Led grading and student queries for one programming assignment.
- Assisted with invigilating and grading all quizzes and exams.

PROJECTS

Open-Vocabulary Semantic Room Navigation | *ROS2, Nav2, YOLO-World, Python* **May 2026**
EE 3002: Junior Design Studio (Robotics), team of 2 · Video demos

- Built a GPU-free object-search robot that classifies rooms from YOLO-World detections to skip wrong rooms early, with semantic memory that avoids re-searching on later missions.
- Moved from Gazebo simulation to a RoboMaster EP Core with a Raspberry Pi 5, offloading inference to a laptop over Wi-Fi via ROS 2.
- Extended into a multi-robot system, *Hive-Nav*, now under review at ICRA 2027.

Scalable Parallel Data Pipeline for Genome-Scale Analysis | *Python, Bash, pandas, bedtools* **May 2026**
CS 5312: Big Data Analytics, team of 3

- Built a four-step pipeline over about 130.8 GB of genomic sequencing data from 927 samples, using bedtools interval arithmetic and chunked, per-sample parallel processing.
- Classified 11,071 genomic regions by methylation state and found that regions overlapping transposable elements are 19.5× more likely to carry transposon-associated methylation.

Evaluation of HTTP Website Communication using NTLMv2 | *Wireshark, HTTP* **Nov. 2025**
CS 5803: Applied Cryptography, team of 3

- Captured a university website's plain-HTTP traffic with Wireshark to assess what a passive eavesdropper could observe.
- Found that NTLMv2 protected the password, but the unencrypted session cookie left users open to session hijacking.

RELEVANT COURSEWORK

- Algorithms
- Applied Cryptography
- Big Data Analytics
- Computer Systems
- Data Science
- Distributed Systems
- Mobile Robotics
- Multi-agent Systems
- Network-Centric Computing
- Network Security
- Operating Systems

TECHNICAL SKILLS

Languages: Python, C, C++, Go
Developer Tools: VS Code, Git, Wireshark
Robotics: ROS2, YOLO, Nav2, SLAM Toolbox
Machine Learning: scikit-learn, pandas, NumPy
Networking: Socket Programming (TCP/UDP), Routing (LS/DV), Threading

ACHIEVEMENTS AND AWARDS

- Dean's Honour List, LUMS (2024–25, 2025–26)
- Full Merit Scholarship in A Levels at Lahore Grammar School Johar Town
- A Levels High Achiever (2 A grades) | O Levels High Achiever (9 A* grades)